# ScrapeBench: Measuring and Improving Face Protection across Web Acquisition Pipelines

Anonymous Author(s)

## ABSTRACT

On a Web increasingly read by generative agents, publishing a portrait and authorizing a model of one's face are hard to keep apart: once an image is uploaded, ordinary acquisition steps silently decide what a customization agent actually receives. Existing anti-customization defenses are validated on pristine files, although an agent may acquire a photograph after a single ordinary operation such as CDN transcoding, feed rendering, detector cropping, restoration, generative re-upload, or purification. We introduce *ScrapeBench*, an acquisition-contract benchmark that evaluates each such operation independently from the same upload, with a matched clean branch, held-out identity references, and an explicit clean-validity gate that prevents image destruction from being credited as protection. The benchmark is paired with a machine-readable consent-manifest direction for compliant agents; it is not a propagation or enforcement mechanism. We further propose ARCP-Web, a reference defense that allocates a bounded, texture-aware perturbation across coarse and detail grids and concentrates its optimization budget on the currently weakest acquisition condition. In a preregistered four-identity feasibility pilot spanning two zero-shot identity readers and a matrix of independent Web interventions, ARCP-Web has the largest paired-gap point estimate in nine of 11 rankable reader–layer cells. The preregistered 40-identity expansion then reorders the field: the targeted defense IDProtector attains the largest paired gap in all 11 rankable cells, GAP-Diff's released defense nearly vanishes under acquisition, and ARCP-Web trails IDProtector at its official default yet is statistically indistinguishable from it under a strict common 8/255 budget in ten of the eleven cells while uploading 5.4 dB more imperceptible images. Acquisition survival is thus a defense-design property that pristine-file evaluation cannot see. We release the harness, intervention registry, per-sample ledgers, the executed 40-identity expansion, and a `face-consent.txt` design direction for future Web systems.

## CCS CONCEPTS

• **Security and privacy** → **Privacy-preserving protocols**; • **Computing methodologies** → *Computer vision*; *Image processing*.

## KEYWORDS

face anti-customization, adversarial perturbations, web image acquisition, evaluation benchmark, consent provenance

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
*WWW '27, May 10–14, 2027, Dublin, Ireland*
© 2027 Association for Computing Machinery.
ACM ISBN 978-x-xxxx-xxxx-x/YY/MM…$15.00
https://doi.org/10.1145/nnnnnnn.nnnnnnn

**ACM Reference Format:**
Anonymous Author(s). 2027. ScrapeBench: Measuring and Improving Face Protection across Web Acquisition Pipelines. In *Proceedings of The ACM Web Conference 2027 (WWW '27).* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

## 1 INTRODUCTION

The Web's old social contract was written for human readers. A person could publish a portrait to be seen without thereby authorizing anyone to construct a generator of their identity. On the agentic Web, that distinction is collapsing: crawlers can collect a few images and invoke DreamBooth [34], InstantID [46], or IP-Adapter-FaceID [58] without further interaction with the subject. Pixel perturbations such as Anti-DreamBooth, GAP-Diff, and ID-Protector attempt to restore agency by making published images unsuitable for customization [20, 41, 60].

**What is brave and new.** We propose that anti-customization be treated as an *acquisition contract* between people, Web infrastructure, and AI agents, not as a property of a single downloaded file. The concrete contribution is a benchmark and reference defense in which ordinary Web interventions are first-class, independently testable contract clauses. This changes the research question from "does noise fool one model?" to "after which legitimate Web operation does a person's refusal cease to be enforceable?" The distinction matters for future browsers, CDNs, agents, and policy: it makes failures attributable and points to where consent must be preserved.

A cumulative funnel—where each stage inherits the damage of all earlier stages—cannot answer that question: it confounds the effect of a JPEG encoder with that of a crop or a diffusion reconstruction. ScrapeBench instead defines seven *single interventions*. T0 is the released image; each of T1–T6 consumes that same T0 source independently. Clean and protected branches use identical identities, operations, prompts, seeds, and scorers. This paired design measures the protection gap attributable to one intervention while separately recording whether the input or generated face remains usable.

Our preregistered feasibility pilot takes the first four valid identities from a manifest fixed before any defense outcome was inspected, with four gallery and two held-out reference images each. The preregistered expansion pool is pre-audited (40 identities, 160 gallery images, 96.25% input-face detection, .787 mean gallery–reference ArcFace similarity), and the identical protocol has been fully executed on it. The completed pilot compares clean uploads, GAP-Diff, MetaCloak, GoodAC, and our ARCP-Web under InstantID and IP-Adapter-FaceID; the expansion adds the targeted defense IDProtector and strict 8/255-projected variants of GAP-Diff and IDProtector. Released checkpoints are used where provided, and MetaCloak and GoodAC are evaluated from their released code with disclosed Web-scenario adaptations. A separate DreamBooth protocol is a gated second-stage experiment.

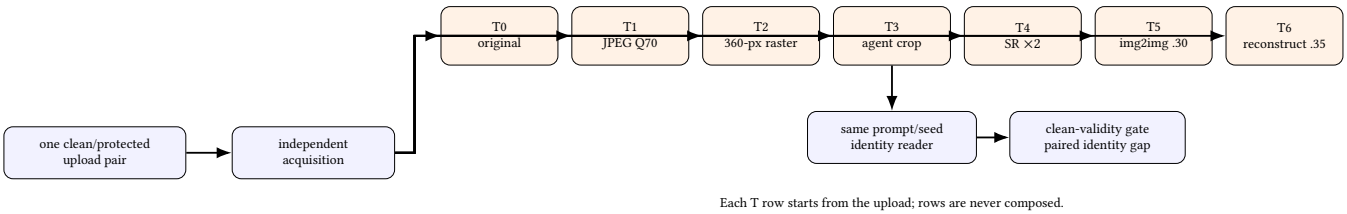



**Figure 1: ScrapeBench motivation and acquisition contract. Conventional pristine-file evaluation observes only T0. ScrapeBench instead forks the same upload into seven independently executed Web interventions, pairs clean and protected branches, and ranks a layer only when its clean attack remains valid. The labels here are the exact registry used in the experiments.**

ARCP-Web is designed for this contract. Its matched-budget evaluation uses a pixel-space 8/255 cap, coarse/detail perturbation grids, texture-aware amplitude allocation, and a differentiable port of InstantID's ArcFace recognizer inside a balanced objective spanning T0–T6. Half of its updates cover all interventions uniformly; the rest target the intervention currently retaining the most identity information. This is a falsifiable design, not a guarantee. We claim an advantage at a layer only if its paired confidence interval supports one.

Our contributions are:

- an acquisition-contract formulation for face consent on the agentic Web;
- a non-cumulative T0–T6 benchmark with matched clean controls, explicit validity gates, and conditional plus end-to-end metrics;
- a preregistered four-identity feasibility pilot and its executed 40-identity expansion, comparing released robustness-oriented and targeted defenses under two zero-shot customization families with paired bootstrap intervals, Holm-corrected direct comparisons, and strict common-budget rows;
- ARCP-Web, whose optimization allocates effort to its weakest acquisition intervention rather than optimizing only the pristine upload; and
- a machine-readable `face-consent.txt` direction for carrying consent when no pixel-only defense can satisfy the contract.

## 2 RELATED WORK

*Customization and identity readers.* Personalized diffusion has matured from latent diffusion backbones deployed at Web scale [33, 40] into a family of subject-driven methods [1, 50]. DreamBooth binds a subject to a rare token from a handful of images [34], textual inversion encodes the subject into a word embedding [8], and multi-concept customization composes several subjects at once [19]. Zero-shot readers remove training entirely: InstantID injects a face embedding through a decoupled cross-attention branch [46], and IP-Adapter-FaceID feeds a face identity embedding as a prompt adapter [58]. Because these readers run in seconds on a single photograph, a scraped portrait is immediately actionable, which is precisely the threat model our benchmark evaluates.

*Adversarial face and identity protection.* Pixel-space defenses perturb images so that customization fails while the image still looks natural. The line began with Anti-DreamBooth's initialization-level attack on DreamBooth fine-tuning [20] and PhotoGuard's encoder attacks [36], and has since specialized along several axes: face-specific robustness [32, 48], face-swap pathways [45], JPEG-robust perturbations [6, 18], real-time identity defense [10], and deepfake-oriented shielding [15]. MetaCloak meta-learns perturbations that survive diverse surrogate losses [25]; GoodAC couples global and local adversarial objectives for anti-customization [54]; GAP-Diff targets the JPEG-transcoded setting directly [60]; IDProtector trains an adversarial noise encoder against zero-shot identity readers [41]; and robust purification-resistant objectives extend protection to diffusion-based restoration [35, 47, 55, 57]. Our pilot benchmarks the three defenses with released code or checkpoints (GAP-Diff, MetaCloak, GoodAC) and contributes ARCP-Web as a reference defense; the remaining variants share the same blind spot we quantify—they are validated on pristine uploads.

*Protection beyond perturbations.* A parallel line changes the generation pathway rather than the pixels. Latent-diffusion unlearning shifts denoising trajectories away from personalized reproduction [5], Anti-Diffusion blocks model modifications [24], and dual-domain anti-personalization attacks the biometric feature space [56]. Immunization against instruction-driven editing and optimization-free diffusion vaccines broaden the attack surface [28, 42], while contrastive training probes the robustness of protective perturbations under latent noise [30]. Identity-unextractable synthesis, cross-image consistency, and dual-layer defenses mix generative and adversarial mechanisms [16, 44, 49], and recent analyses quantify how fragile adversarial defenses remain under distribution shift [2]. Extension to video protection shows the same acquisition problem appearing in temporal generators [11, 13].

*Unlearnable examples and data poisoning.* Availability attacks predate customization defenses: unlearnable examples degrade training from the data side [12], with robust variants [7] and autoregressive perturbations [37]. Augmentation-aware analyses then showed that benign preprocessing defeats many such poisons [17, 31]. The modern toolbox spans provably unlearnable data [43], temporal unlearnable videos [51], trigger-free poisoning [14], adversarial mislabeling [52], face-privacy unlearnable synthesis [53], detection of convolution-based poisons [23], augmentation-shielded designs [9], and systematizations of the field [22, 59]. Artist protection shares the adversarial mechanics [38] and inherits the same robustness questions raised by Fawkes-style cloaking [39].

*Evaluation, recognition, and consent.* Large-scale benchmarks standardized face-forgery evaluation under realistic degradations [27, 29], but no benchmark fixes the Web acquisition channel

**Table 1: Independent acquisition conditions. Every row starts from the same T0 image; T1–T6 are never composed in the primary analysis.**

| Layer | Intervention | Web actor or behavior |
|---|---|---|
| T0 | original protected upload | publisher |
| T1 | JPEG Q70, 4:2:0, 512 px | CDN transcode |
| T2 | 360-px CSS raster, DPR 1 | feed/thumbnail render |
| T3 | detector crop, margin .35, WebP 80 | screenshot agent |
| T4 | Real-ESRGAN ×2 | restoration service |
| T5 | SD1.5 img2img, strength .30 | generative re-upload |
| T6 | prompt-free SD1.5 reconstruction, strength .35 | purification |

for anti-customization. Recognition metrics ground our measurements in face embeddings [4], evaluated on in-the-wild face collections [26]. On the governance side, empirical studies document how rarely Web-scraped corpora carry consent signals [21], and content-provenance specifications provide a machine-readable channel for them [3]. ScrapeBench connects these threads: it fixes the acquisition contract as the unit of measurement and pairs pixel-level robustness with consent metadata that survives when perturbations do not.

# 3 THE SCRAPEBENCH ACQUISITION CONTRACT

## 3.1 Threat model

Let subject $s$ publish gallery $X_s$ and defense $D$ release $\widetilde{X}_s = D(X_s)$. An agent obtains either branch through exactly one acquisition intervention $\phi_k$ and passes it to customization pipeline $P$. The held-out reference set $R_s$ is never protected, transformed, or supplied to $P$; it is used only for evaluation. The benchmark pairs $P(\phi_k(X_s))$ with $P(\phi_k(\widetilde{X}_s))$ using the same prompt and seed. Thus a platform operation cannot be mistaken for protection.

## 3.2 Seven independent interventions

The intervention set $\Phi = \{\phi_0, \ldots, \phi_6\}$ is a flat matrix, not a funnel: each $\phi_k$ consumes T0 and never the output of another row.

T1 matches GAP-Diff's primary JPEG setting and sits at the pessimistic end of the CDN qualities we measured (GitHub≈75, WordPress≈80). T2 approximates a common feed card; T3 represents an agent that sees rendered pixels rather than the origin file; T4–T6 test increasingly model-mediated acquisition. T6 is explicitly a prompt-free reconstruction proxy, not an implementation claimed to be the official DiffPure algorithm. Each layer has a usability check on its input face.

## 3.3 Metrics and validity gates

Let $e(y, R_s)$ be the maximum ArcFace cosine similarity between generated image $y$ and valid references of $s$. We report conditional identity similarity (ISM) over detected generated faces and end-to-end ISM, where detection failures receive zero. Face-detection failure rate (FDFR) is reported separately, so an undetectable image is never silently credited as identity-specific protection. For defense $D$, attack $P$, and layer $k$, the paired protection gap is

$$\Delta_{k,D,P} = \mathbb{E}_s[\mathrm{ISM}_{k,\mathrm{clean},P}(s) - \mathrm{ISM}_{k,D,P}(s)]. \tag{1}$$

Higher $\Delta$ is better. Retention ratio is $\mathrm{RR}_{k,D,P} = \Delta_{k,D,P}/\Delta_{0,D,P}$, accompanied by a paired identity-bootstrap interval and the absolute change $\Delta_k - \Delta_0$. Protected uploads are audited by their realized per-image $L_\infty$ after 8-bit quantization and by PSNR. A layer is rankable only when the matched clean attack detects at least 50% of outputs and retains at least 50% of its clean-T0 end-to-end ISM. A defense is called effective only when that gate passes and the 95% interval for $\Delta$ excludes zero.

## 3.4 Beyond pixels: consent carried by the Web

No bounded perturbation can guarantee survival under arbitrary reconstruction. We therefore sketch `/.well-known/face-consent.txt`, a machine-readable manifest mapping image patterns and subject hashes to permitted uses, expiry, revocation, and optional signatures. In its present form this is only a voluntary preference and audit hook for compliant agents. It neither proves that an uploader can authorize every depicted subject, locates a source after arbitrary copying, nor constrains a malicious agent. ScrapeBench supplies the empirical test for pixel defenses; origin binding, subject verification, cross-domain propagation, conflict resolution, and enforcement remain separate Web-systems research problems.

# 4 ARCP-WEB: OPTIMIZING THE WEAKEST CONTRACT CLAUSE

ARCP-Web is a reference defense for the independent intervention matrix. It is not trained on outputs from a seven-stage cascade. For image $x$, a coarse grid $z_c \in [-1, 1]^{128\times128\times3}$ and a full-resolution detail grid $z_d$ are combined inside a fixed texture map $A(x)$:

$$\widetilde{x} = \Pi_{x,\epsilon_{\max}}\big(x + A(x) \odot (.45U(z_c) + .55z_d)\big), \tag{2}$$

where the fixed texture map is $A(x) = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})\widehat{t}(x)$. $\widehat{t}$ is smoothed Sobel energy normalized at its 95th percentile, $\epsilon_{\min} = 3/255$, and $\epsilon_{\max} = 8/255$ in pixel space; $U$ denotes element-wise projection of the update onto the unit ball before scaling by the texture amplitude. The coarse branch improves survival while the detail branch retains capacity against the T0 reader; projection is repeated after 8-bit PNG quantization.

For each layer we define a differentiable training proxy $q_k$: identity, JPEG-Q70, 360-pixel resampling, jittered face crop, a super-resolution down/up-sharpen proxy, a mild reconstruction bottleneck, and a stronger low-pass reconstruction bottleneck. The exact Real-ESRGAN and diffusion models remain unseen evaluation transforms. Given the clean identity centroid $c_s$, ARCP-Web maximizes

$$\begin{aligned}\mathcal{J}_k = {} & \mathcal{L}_{\mathrm{denoise}}(q_k(\widetilde{x})) \\ & - \lambda\big([s_k^D - m]_+ + [s_k^D - s_k^C + \rho]_+\big) + \gamma\,\|q_k(\widetilde{x}) - q_k(x)\|_2^2.\end{aligned} \tag{3}$$

Here $s_k^D$ and $s_k^C$ are protected and matched-clean similarities to $c_s$. The identity objective ensembles FaceNet with a differentiable port of InstantID's released antelopev2 ArcFace recognizer [4]. Face landmarks are detected once on the clean input and held fixed, leaving the recognition path differentiable without optimizing through a detector. The first term poisons training-based customization, the second repels the identity readers and requires paired gap $\rho = .25$, and the survival term maximizes the protected-to-clean output difference under the shared proxy, i.e., it rewards a perturbation that

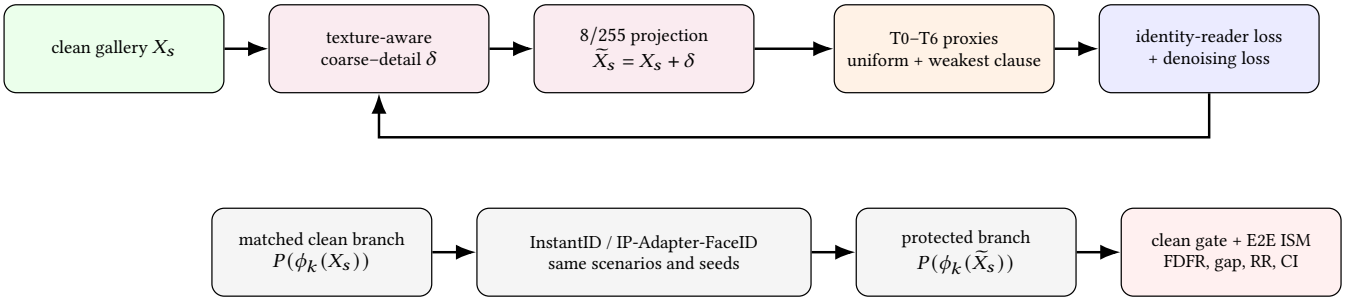



**Figure 2: ARCP-Web optimization (top) and paired ScrapeBench evaluation (bottom). ARCP-Web is a self-contained clean-to-protected method. Its adaptive scheduler alternates coverage of every registered intervention with the currently weakest paired clause. Validity is decided separately for each reader and layer; no fixed subset of T4–T6 is declared invalid in advance.**

remains present after the same stochastic operation is applied to both branches.

The scheduler prevents any single layer from dominating. Even steps cycle through T0–T6, guaranteeing equal coverage. Odd steps select the layer with the smallest exponential-moving-average paired clean-minus-protected gap—the current weakest clause. Optimization uses 600 signed-gradient updates, $\lambda = 1.20$, $m = .10$, and $\gamma = .35$; all three weights were fixed from the design targets before the pilot was scored, and the preregistered 40-identity expansion provides the held-out check. A corrected implementation converts ARCP tensors from $[-1, 1]$ to $[0, 1]$ before differentiable JPEG; without that conversion the JPEG proxy clips its input and does not train the intended condition.

This construction aims for a positive paired gap at every T0–T6 layer. It does not make that outcome true by definition: only held-out InstantID, IP-Adapter-FaceID, and DreamBooth measurements can establish it. Failed layers remain in the table and determine the next design iteration.

# 5 EXPERIMENTAL PROTOCOL

## 5.1 Four-identity feasibility pilot

The feasibility pilot preregisters four identities, each with four gallery images and two held-out references. They are the first four valid entries of a manifest fixed before any defense outcome was inspected. The expansion pool contains 40 audited identities (35 CelebA [26] and five VGGFace2 identities from the GAP-Diff artifact): across its 160 gallery images, input face detection is 96.25% and mean gallery–reference ArcFace similarity is .787. The pilot validates the protocol end to end and establishes effect direction; it cannot support population claims.

## 5.2 Defenses and attacks

The primary rows are clean, GAP-Diff [60], MetaCloak [25], GoodAC [54], and ARCP-Web. GAP-Diff uses its released checkpoint and preprocessing. MetaCloak and GoodAC are evaluated from their released implementations and published hyperparameters, with every Web40 adaptation (input split and identity-neutral prompt) recorded in the released configuration. A method enters the table only after protection completeness, realized pixel budget, and clean-attack validity checks pass. Methods whose released default exceeds 8/255 (GAP-Diff reaches 9/255) are additionally reported under their official default, separately from the common-budget comparison.

**Table 2: Two gates answer different questions and are never substituted.**

| Gate | Threshold | Consequence |
|---|---|---|
| Reader smoke | det.≥ 75%, cond. ISM≥ .25 | run full grid |
| Layer rankability | det.≥ 50%, retention≥ 50% | permit ranking |
| Defense effect | 95% CI for paired gap excludes 0 | call effective |

Stage A evaluates InstantID and IP-Adapter-FaceID, with 30 denoising steps, guidance 5, and four outputs per cell—one for each fixed Web scenario: social avatar, smartphone selfie, low-light indoor social photograph, and outdoor travel photograph. Before the full grid, a four-identity clean-T0 smoke test must attain mean conditional ISM at least .25 and at least 75% valid generated faces for each reader; layers whose production clean branch later falls below the 50% rankability threshold are excluded from ranking regardless of the smoke result. Stage B is a gated second-stage experiment (SD2.1 DreamBooth, 1,000 steps, prior preservation with 200 class images, text-encoder training) that runs only after the zero-shot pilot shows a useful signal. Zero-shot and fine-tuning results are never pooled.

The Web40 expansion preregisters 40-identity primary rows for Clean, GAP-Diff, IDProtector, and ARCP-Web under both readers and every independent T0–T6 intervention. Official-default rows retain released outputs; additional GAP-Diff and IDProtector rows are projected against their clean inputs to a strict realized 8/255 ceiling. Six one-factor ARCP ablations (uniform rather than weakest-clause scheduling, no texture map, no coarse branch, no detail branch, no identity objective, and no denoising objective) use the fixed first four identities as a mechanism diagnostic. MetaCloak and GoodAC are not rerun at 40 identities; their released defenses remain in the four-identity pilot comparison. Two additional seed replicas of the pilot (seed blocks 20261019 and 20261020, stage A) support the seed-stability analysis in Section 6. The expansion is accepted only if all expected sample ledgers exist; tables are generated from those ledgers, not transcribed manually.

### 5.3 Execution and reproducibility contract

Every cell is deterministic from its identity, layer, reader, prompt, and seed. Generated images and a JSONL sample ledger are saved, and summary rows are append-only. Execution is gate-ordered: the clean smoke test precedes protection generation, and attack evaluation is launched only after both gates pass—a failed gate blocks downstream jobs rather than yielding a plausible-looking table built from an invalid attack.

## 6 RESULTS: FEASIBILITY PILOT AND THE 40-IDENTITY EXPANSION

We report the completed `web4_validation` pilot. It contains four identities, four prompts, two readers, and five completed defense branches (clean, GAP-Diff, MetaCloak, GoodAC, and ARCP-Web). All values below are recomputed from 280 per-cell JSONL ledgers. Lower end-to-end ISM is better; higher paired clean-minus-protected gap Δ is better.

*Reader validity.* InstantID passes the clean gate at T0–T3 and T5. Its T4 branch detects only 25% of outputs (E2E ISM .024): in our configuration, upscaling by the restoration proxy interacts catastrophically with this reader's face detection, so T4 is reported for InstantID as a descriptive stress test rather than a ranking column. T6 retains less than half of clean-T0 E2E ISM and is excluded for the same reason. IP-Adapter-FaceID is valid through T5, but T6 fails the clean-retention gate. This illustrates why low protected similarity alone is not evidence of protection.

*ARCP effect direction.* Figure 3 visualizes every rankable paired gap with its identity-bootstrap interval; shaded columns are gate-invalid stress tests. Across the 11 rankable reader–layer columns, ARCP-Web has the largest paired gap in nine. For InstantID, ARCP leads at T0–T3 with gaps .216, .199, .177, and .121; their identity-bootstrap 95% intervals are [.101,.328], [.077,.336], [.057,.246], and [.044,.197]. MetaCloak instead leads T5 (.118, [.016,.179]), slightly above GoodAC (.108). For IP-Adapter-FaceID, ARCP leads T0–T4 with gaps .148, .140, .180, .165, and .073; all five intervals exclude zero. GoodAC leads T5 with .069, although its interval [−.032, .168] crosses zero. Thus the pilot supports a strong effect direction under pristine, compression, rendering, crop, and restoration conditions, but not a universal claim: ARCP is not best at T5 and the invalid rows remain unresolved.

*Retention and detection.* Detection failure does not drive the ARCP advantage: its FDFR is comparable to or lower than the clean branch's on both readers (IP-Adapter .125–.188 at T0–T3 against clean's .062; InstantID .312 at T0–T3 against clean's .375–.438). ARCP's InstantID RR is .918, .818, and .558 at valid T1–T3; its IP-Adapter RR is .951, 1.218, 1.116, and .494 at T1–T4, and RR at T5 is only .151 and .321. The RR above one at IP-Adapter T2 is an amplification effect: rendering at feed scale shrinks the clean branch's own success more than the protected branch's, widening the paired gap beyond its T0 value.

At the same realized 8/255 ceiling as GoodAC, ARCP has 4.35 dB higher mean PSNR. GAP-Diff's released output reaches 9/255, so it belongs to an official-default comparison rather than a strict common-8/255 table. Figure 4 summarizes the resulting trade-off: ARCP-Web is the only defense that simultaneously attains the strictest realized budget and the largest mean paired gap over rankable cells.

These results are a feasibility gate, not a population estimate. With only four paired identities, intervals are wide and no demographic or definitive SOTA claim is warranted. The identical protocol now turns to the pre-audited 40-identity pool.

### 6.1 Forty-identity expansion

The expansion executes the identical preregistered protocol on the full 40-identity pool (Tables 7 and 8). Its defense set follows the expansion preregistration: Clean, GAP-Diff at its official 9/255 default, the targeted zero-shot defense IDProtector [41], and ARCP-Web, with strict 8/255-projected variants of GAP-Diff and IDProtector reported separately; MetaCloak and GoodAC remain in the four-identity pilot comparison. Rankability is now reader-specific: for InstantID, SR×2, img2img, and both purification strengths fail the clean gate (clean ISM .059–.209 against a .475 clean-T0 ceiling), so T0–T3 are rankable, while for IP-Adapter-FaceID the same operations leave the clean attack largely intact and T0–T6 remain rankable except the strength-.50 purification. Eleven reader–layer cells enter the ranking, each with identity-bootstrap intervals.

The expansion reorders the pilot's provisional ranking. IDProtector attains the largest paired gap in all 11 rankable cells: InstantID gaps are .417, .392, .386, and .406 at T0–T3, and IP-Adapter-FaceID gaps are .216, .188, .191, .171, .177, .060, and .077 at T0–T6. GAP-Diff's released defense nearly vanishes under acquisition (.023 and .054 at T0; at most .054 in any rankable cell). ARCP-Web is second in ten of the eleven cells and third at IP-Adapter-FaceID T5. Against IDProtector at official default, the paired identity-resampled difference is significantly negative for ARCP-Web in nine of the eleven cells, e.g., −0.115 with interval [−0.158, −0.072] at InstantID T0 (Holm-corrected $p = .0022$).

Under the strict common 8/255 ceiling the ordering largely closes: projecting IDProtector to the same realized budget as ARCP-Web leaves the two statistically indistinguishable in ten of the eleven rankable cells (InstantID T0: −0.087 [−0.129, −0.046], Holm $p = .058$; IP-Adapter-FaceID T0: −0.006 [−0.033, 0.019]), with InstantID T3 the single exception (−0.111 [−0.153, −0.071], $p = .0013$). At that same ceiling ARCP-Web's mean upload PSNR is 37.71 dB against IDProtector's 32.34 dB—a 5.4 dB imperceptibility advantage at statistically comparable protection. Official-default and common-budget rows are never pooled.

A large share of IDProtector's protected outputs defeats face detection itself: FDFR is .606–.619 on InstantID T0–T3 and .500–.512 on IP-Adapter-FaceID T0–T3, against .231–.312 and .144–.156 for ARCP-Web. Its gaps are therefore real but partially reflect "no face" outcomes that the ledger reports explicitly rather than a hidden zero, and its usability cost is mirrored by the PSNR gap above.

Six one-factor ARCP-Web ablations on the fixed four identities are mechanism diagnostics, not rankings: every Holm-corrected $p$ equals one at $n = 4$. Directionally, the identity objective is the only component with a consistent effect—removing it costs .102–.216 of paired advantage across the rankable InstantID cells and .083–.159 across IP-Adapter-FaceID T0–T4, with no consistent direction at the generative tiers—the weakest-clause scheduler beats uniform

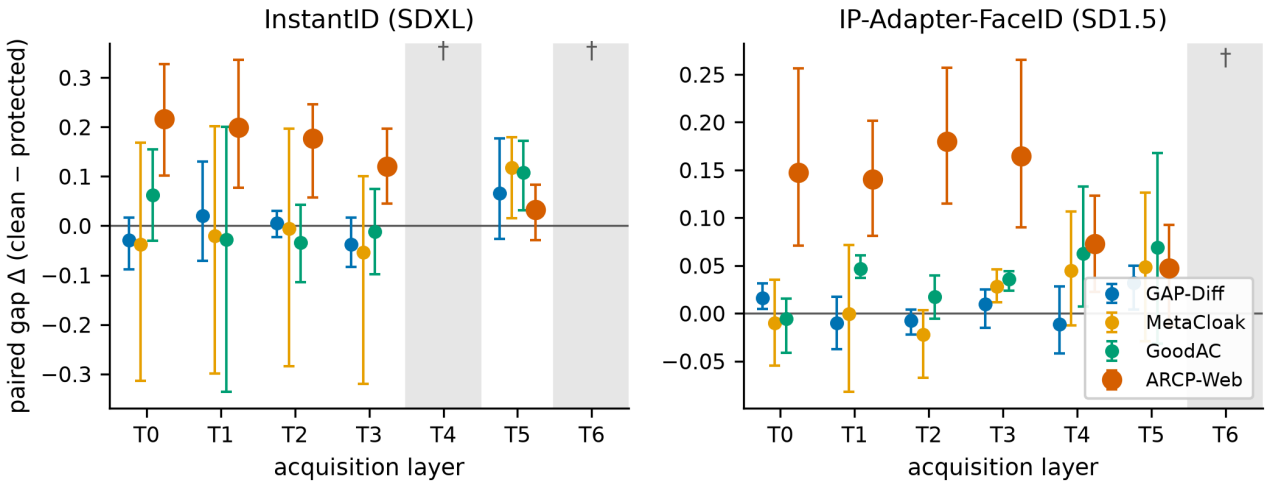



**Figure 3: Paired protection gap Δ (clean minus protected) with identity-bootstrap 95% intervals for every rankable reader–layer cell. Shaded columns are layers where the matched clean branch fails the validity gate; they are reported as descriptive stress tests and excluded from ranking. ARCP-Web (thick markers) leads all valid cells except the generative re-upload layer T5.**

**Table 3: Four-scenario comparison under the InstantID reader after JPEG compression at quality 70 (T1), averaged over the four preregistered identities. FDFR is reported as a percentage; conditional ISM averages only detected faces, whereas end-to-end (E2E) ISM assigns zero to a detection failure. The paired gap is $\Delta$ = E2E $\text{ISM}_{\text{clean}}$ − E2E $\text{ISM}_{\text{defense}}$. Lower ISM and higher $\Delta$ indicate stronger protection. FDFR is diagnostic rather than a stand-alone success criterion. Bold marks the strongest non-tied defense result; the avatar row is not ranked because all four defenses produce 100% detection failure.**

| | Social-media avatar | | | | Ordinary smartphone selfie | | | |
|---|---|---|---|---|---|---|---|---|
| Method | FDFR ↑ | Cond. ISM ↓ | E2E ISM ↓ | Δ ↑ | FDFR ↑ | Cond. ISM ↓ | E2E ISM ↓ | Δ ↑ |
| No defense | 75.0 | 0.583 | 0.146 | – | 25.0 | 0.495 | 0.371 | – |
| GAP-Diff | 100.0 | – | 0.000 | 0.146 | 25.0 | 0.494 | 0.370 | 0.000 |
| MetaCloak | 100.0 | – | 0.000 | 0.146 | 0.0 | 0.494 | 0.494 | −0.123 |
| GoodAC | 100.0 | – | 0.000 | 0.146 | 0.0 | 0.479 | 0.479 | −0.108 |
| ARCP-Web | 100.0 | – | 0.000 | 0.146 | 0.0 | **0.129** | **0.129** | **0.242** |
| | **Low-light indoor social photo** | | | | **Outdoor travel photo with complex background** | | | |
| Method | FDFR ↑ | Cond. ISM ↓ | E2E ISM ↓ | Δ ↑ | FDFR ↑ | Cond. ISM ↓ | E2E ISM ↓ | Δ ↑ |
| No defense | 25.0 | 0.519 | 0.389 | – | 25.0 | 0.510 | 0.382 | – |
| GAP-Diff | 0.0 | 0.413 | 0.413 | −0.024 | 0.0 | 0.422 | 0.422 | −0.040 |
| MetaCloak | 25.0 | 0.499 | 0.374 | 0.015 | 0.0 | 0.499 | 0.499 | −0.117 |
| GoodAC | 0.0 | 0.451 | 0.451 | −0.062 | 0.0 | 0.470 | 0.470 | −0.087 |
| ARCP-Web | 25.0 | **0.248** | **0.186** | **0.203** | 0.0 | **0.178** | **0.178** | **0.204** |

scheduling in ten of eleven cells (mean +0.019), and the texture map trades raw ISM reduction for image quality in all fourteen ablation cells. Two additional seed replicas of the pilot (16 reader–layer cells each) rank IDProtector above GAP-Diff in 15 of 16 seed-cells and above ARCP-Web in 12 of 16; every disagreement lies inside the pilot's wide identity-bootstrap intervals and resolves in IDProtector's favor at $n = 40$.

## 7 DISCUSSION

The acquisition-contract view connects an algorithmic privacy question to Web architecture. A failure at T1 implicates image-serving infrastructure; a failure at T3 implicates what an agent observes after browser rendering; a failure at T5 or T6 shows that a pixel-only refusal cannot survive generative mediation. Independent layers make those diagnoses possible, while a cascade would not.

The four-identity feasibility pilot is intentionally a fast falsification test; the preregistered 40-identity expansion then resolves its provisional ranking. Across the 11 rankable reader–layer cells at $n = 40$, the targeted defense IDProtector holds the largest paired gap in every cell, ARCP-Web follows in ten and trails IDProtector significantly at the official default in nine, and the two become statistically indistinguishable under a strict common 8/255 budget

**Table 4: End-to-end identity similarity (E2E ISM; lower is better). Detection failures receive zero and are also reported separately as FDFR. † marks a layer where the clean validity gate fails; such columns are descriptive stress tests and are not used to rank defenses. Leading zeros are shown to make the scale explicit.**

| Reader | Defense | T0 | T1 | T2 | T3 | T4 | T5 | T6 |
|---|---|---|---|---|---|---|---|---|
| InstantID | Clean | 0.318 | 0.322 | 0.288 | 0.304 | 0.024† | 0.189 | 0.112† |
| | GAP-Diff | 0.347 | 0.301 | 0.283 | 0.341 | 0.007† | 0.122 | 0.094† |
| | MetaCloak | 0.356 | 0.342 | 0.293 | 0.357 | −0.001† | 0.071 | 0.008† |
| | GoodAC | 0.255 | 0.350 | 0.321 | 0.316 | 0.021† | 0.081 | 0.021† |
| | ARCP-Web | 0.101 | 0.123 | 0.111 | 0.183 | 0.034† | 0.156 | 0.054† |
| IP-Adapter | Clean | 0.408 | 0.413 | 0.411 | 0.421 | 0.451 | 0.259 | 0.178† |
| | GAP-Diff | 0.393 | 0.423 | 0.419 | 0.412 | 0.462 | 0.227 | 0.140† |
| | MetaCloak | 0.418 | 0.413 | 0.434 | 0.393 | 0.406 | 0.210 | 0.122† |
| | GoodAC | 0.414 | 0.366 | 0.394 | 0.386 | 0.388 | 0.190 | 0.082† |
| | ARCP-Web | 0.261 | 0.272 | 0.232 | 0.257 | 0.378 | 0.212 | 0.126† |

**Table 5: Paired protection gap $\Delta = \mathrm{ISM}_{clean} - \mathrm{ISM}_{defense}$ (higher is better). This is the ranking metric because every defense is compared with the same identity's clean branch. Bold is the largest gap in each valid reader–layer column. Full identity-bootstrap intervals are reported in the released CSV and discussed below.**

| Reader | Defense | T0 | T1 | T2 | T3 | T4 | T5 | T6 |
|---|---|---|---|---|---|---|---|---|
| InstantID | GAP-Diff | −0.029 | 0.021 | 0.005 | −0.037 | 0.017† | 0.066 | 0.018† |
| | MetaCloak | −0.038 | −0.020 | −0.005 | −0.053 | 0.025† | **0.118** | 0.104† |
| | GoodAC | 0.063 | −0.028 | −0.033 | −0.012 | 0.003† | 0.108 | 0.090† |
| | ARCP-Web | **0.216** | **0.199** | **0.177** | **0.121** | −0.010† | 0.033 | 0.058† |
| IP-Adapter | GAP-Diff | 0.016 | −0.010 | −0.007 | 0.010 | −0.011 | 0.032 | 0.038† |
| | MetaCloak | −0.010 | −0.000 | −0.022 | 0.028 | 0.045 | 0.049 | 0.056† |
| | GoodAC | −0.006 | 0.047 | 0.018 | 0.036 | 0.062 | **0.069** | 0.095† |
| | ARCP-Web | **0.148** | **0.140** | **0.180** | **0.165** | **0.073** | 0.047 | 0.051† |

**Table 6: Realized upload distortion over the four fixed identities.**

| Defense | Max. 8-bit $L_\infty$ | Mean PSNR (dB) |
|---|---|---|
| GAP-Diff | 9/255 | 31.68 |
| MetaCloak | 11/255 | 29.76 |
| GoodAC | 8/255 | 33.45 |
| ARCP-Web | 8/255 | 37.80 |

in ten of eleven cells—with ARCP-Web uploading 5.4 dB more imperceptible images. GAP-Diff, the strongest pristine-file defense in earlier comparisons, nearly vanishes under acquisition, confirming that pristine-file rank order is not predictive of acquisition survival. The generative tiers T5–T6 remain the weakest for every defense, and the pilot's wide $n = 4$ intervals explain why two seed replicas still disagree with the final ordering in a minority of cells. Real platform upload/download traces, additional identity distributions, and adaptive attacks remain necessary. ARCP-Web's training proxies approximate T4–T6 rather than differentiating through their exact evaluation implementations, so transfer is an empirical risk. Threats to validity also include the single generation seed per scenario in the expansion, detector dependence of all reported metrics, and port fidelity for the two released-code defenses. Most importantly, robustness of a perturbation does not create a legal or social duty for an agent to respect it; that is why the technical benchmark is paired with a consent-manifest design direction.

## 8 CONCLUSION

ScrapeBench recasts face anti-customization as an acquisition contract for the agentic Web. Its seven layers are independent interventions from one common T0, not a cumulative funnel. The preregistered four-identity feasibility pilot compares recent released-code defenses and ARCP-Web under two zero-shot customization readers, with hard validity gates and paired reporting, and the identical protocol has now been executed on the full pre-audited 40-identity pool. The expansion reorders the field: IDProtector holds the largest paired gap in all 11 rankable reader–layer cells, ARCP-Web follows in ten and matches IDProtector under a strict common 8/255 budget in ten of eleven cells while uploading 5.4 dB more imperceptible images, and GAP-Diff's protection collapses under acquisition. The generative re-upload tier T5 and the purification tier T6 remain open for every defense—exactly the boundary at which consent metadata, not pixels, must carry the contract. This combination

**Table 7: Forty-identity results for InstantID. Lower E2E ISM/FDFR-adjusted identity similarity is better; higher paired gap is better. Parentheses give the identity-bootstrap 95% CI. A dagger marks a layer that fails the preregistered clean rankability gate.**

| Layer | Defense | IDs | E2E ISM↓ | Paired gap↑ [95% CI] |
|---|---|---|---|---|
| T0 | Clean | 40 | 0.475 | – |
| T0 | GAP-Diff | 40 | 0.452 | 0.023 [-0.025,0.065] |
| T0 | IDProtector | 40 | 0.059 | 0.417 [0.352,0.479] |
| T0 | ARCP-Web | 40 | 0.174 | 0.302 [0.249,0.356] |
| T1 | Clean | 40 | 0.455 | – |
| T1 | GAP-Diff | 40 | 0.443 | 0.012 [-0.035,0.049] |
| T1 | IDProtector | 40 | 0.063 | 0.392 [0.333,0.451] |
| T1 | ARCP-Web | 40 | 0.186 | 0.269 [0.214,0.322] |
| T2 | Clean | 40 | 0.445 | – |
| T2 | GAP-Diff | 40 | 0.427 | 0.018 [-0.026,0.058] |
| T2 | IDProtector | 40 | 0.058 | 0.386 [0.324,0.443] |
| T2 | ARCP-Web | 40 | 0.172 | 0.273 [0.220,0.325] |
| T3 | Clean | 40 | 0.465 | – |
| T3 | GAP-Diff | 40 | 0.452 | 0.013 [-0.028,0.056] |
| T3 | IDProtector | 40 | 0.059 | 0.406 [0.344,0.467] |
| T3 | ARCP-Web | 40 | 0.214 | 0.251 [0.190,0.310] |
| T4† | Clean | 40 | 0.059 | – |
| T4† | GAP-Diff | 40 | 0.064 | -0.005 [-0.023,0.013] |
| T4† | IDProtector | 40 | 0.026 | 0.032 [0.017,0.048] |
| T4† | ARCP-Web | 40 | 0.013 | 0.046 [0.027,0.065] |
| T5† | Clean | 40 | 0.209 | – |
| T5† | GAP-Diff | 40 | 0.134 | 0.075 [0.042,0.107] |
| T5† | IDProtector | 40 | 0.069 | 0.140 [0.105,0.175] |
| T5† | ARCP-Web | 40 | 0.152 | 0.057 [0.022,0.093] |
| T6† | Clean | 40 | 0.206 | – |
| T6† | GAP-Diff | 40 | 0.162 | 0.044 [0.020,0.068] |
| T6† | IDProtector | 40 | 0.103 | 0.103 [0.077,0.128] |
| T6† | ARCP-Web | 40 | 0.145 | 0.061 [0.043,0.079] |

**Table 8: Forty-identity results for IP-Adapter-FaceID. Lower E2E ISM/FDFR-adjusted identity similarity is better; higher paired gap is better. Parentheses give the identity-bootstrap 95% CI. A dagger marks a layer that fails the preregistered clean rankability gate.**

| Layer | Defense | IDs | E2E ISM↓ | Paired gap↑ [95% CI] |
|---|---|---|---|---|
| T0 | Clean | 40 | 0.315 | – |
| T0 | GAP-Diff | 40 | 0.261 | 0.054 [0.028,0.083] |
| T0 | IDProtector | 40 | 0.099 | 0.216 [0.174,0.258] |
| T0 | ARCP-Web | 40 | 0.199 | 0.116 [0.070,0.164] |
| T1 | Clean | 40 | 0.293 | – |
| T1 | GAP-Diff | 40 | 0.265 | 0.028 [0.002,0.056] |
| T1 | IDProtector | 40 | 0.104 | 0.188 [0.149,0.228] |
| T1 | ARCP-Web | 40 | 0.212 | 0.080 [0.039,0.122] |
| T2 | Clean | 40 | 0.296 | – |
| T2 | GAP-Diff | 40 | 0.256 | 0.040 [0.011,0.070] |
| T2 | IDProtector | 40 | 0.104 | 0.191 [0.150,0.234] |
| T2 | ARCP-Web | 40 | 0.211 | 0.084 [0.037,0.130] |
| T3 | Clean | 40 | 0.285 | – |
| T3 | GAP-Diff | 40 | 0.262 | 0.024 [-0.003,0.052] |
| T3 | IDProtector | 40 | 0.114 | 0.171 [0.129,0.214] |
| T3 | ARCP-Web | 40 | 0.222 | 0.063 [0.022,0.104] |
| T4 | Clean | 40 | 0.316 | – |
| T4 | GAP-Diff | 40 | 0.289 | 0.026 [0.000,0.057] |
| T4 | IDProtector | 40 | 0.139 | 0.177 [0.142,0.213] |
| T4 | ARCP-Web | 40 | 0.248 | 0.068 [0.034,0.102] |
| T5 | Clean | 40 | 0.181 | – |
| T5 | GAP-Diff | 40 | 0.142 | 0.038 [0.020,0.057] |
| T5 | IDProtector | 40 | 0.121 | 0.060 [0.036,0.082] |
| T5 | ARCP-Web | 40 | 0.158 | 0.023 [-0.002,0.046] |
| T6 | Clean | 40 | 0.229 | – |
| T6 | GAP-Diff | 40 | 0.182 | 0.048 [0.029,0.065] |
| T6 | IDProtector | 40 | 0.152 | 0.077 [0.057,0.097] |
| T6 | ARCP-Web | 40 | 0.158 | 0.071 [0.053,0.089] |

of falsifiable pixel tests and durable Web consent metadata offers a concrete path from laboratory evaluation to enforceable human agency on the agentic Web.

## REFERENCES

[1] Pu Cao, Feng Zhou, Qing Song, and Lu Yang. 2026. Controllable Generation With Text-to-Image Diffusion Models: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 48, 4 (2026), 4771–4791. https://doi.org/10.1109/TPAMI.2025.3646548

[2] Zhen Chen, Yi Zhang, Xiangyu Yin, Chengxuan Qin, Xingyu Zhao, Xiaowei Huang, and Wenjie Ruan. 2025. Fragile by Design: On the Limits of Adversarial

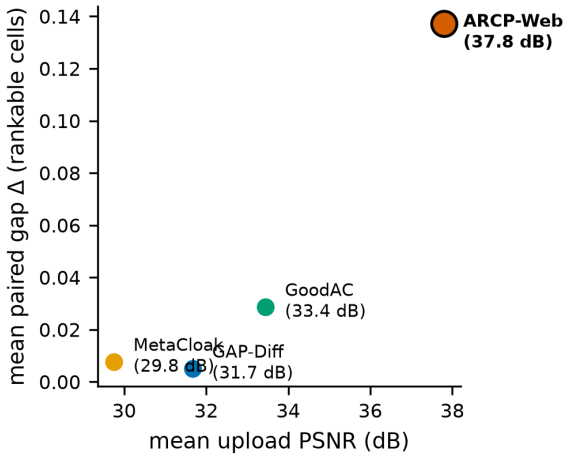



**Figure 4: Protection versus upload quality in the four-identity official-default pilot. Mean paired gap over rankable cells (both readers) is plotted against realized upload PSNR. Budgets are heterogeneous here (Table 6); this plot is not a strict common-budget claim.**


Defenses in Personalized Generation. arXiv:2511.10382. arXiv:2511.10382 http://arxiv.org/abs/2511.10382v1

[3] Coalition for Content Provenance and Authenticity. 2026. C2PA Technical Specification: Content Credentials. C2PA Technical Specification, https://c2pa.org/specifications/.

[4] Jiankang Deng, Jia Guo, Jing Yang, Niannan Xue, Irene Kotsia, and Stefanos Zafeiriou. 2022. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 10 (2022), 5962–5979. https://doi.org/10.1109/TPAMI.2021.3087709 arXiv:1801.07698

[5] Naresh Kumar Devulapally, Shruti Agarwal, Tejas Gokhale, and Vishnu Suresh Lokhande. 2025. Latent Diffusion Unlearning: Protecting Against Unauthorized Personalization Through Trajectory Shifted Perturbations. In *Proceedings of the 33rd ACM International Conference on Multimedia*. ACM. https://doi.org/10.1145/3746027.3755112

[6] Tanjim Rahaman Fardin, S M Zunaid Alam, Mahadi Hasan Fahim, and Md Faysal Mahfuz. 2026. MetaCloak-JPEG: JPEG-Robust Adversarial Perturbation for Preventing Unauthorized DreamBooth-Based Deepfake Generation. arXiv:2604.18537. arXiv:2604.18537 http://arxiv.org/abs/2604.18537v1

[7] Shaopeng Fu, Fengxiang He, Yang Liu, Li Shen, and Dacheng Tao. 2023. Robust Unlearnable Examples: Protecting Data Against Adversarial Learning. In *International Conference on Learning Representations (ICLR)*.

[8] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H. Bermano, Gal Chechik, and Daniel Cohen-Or. 2023. An Image is Worth One Word: Customizing Text-to-Image Generation using Textual Inversion. In *International Conference on Learning Representations (ICLR)*.

[9] Xueluan Gong, Yuji Wang, Yanjiao Chen, Haocheng Dong, Yiming Li, Mengyuan Sun, Shuaike Li, and Qian Wang. 2026. Armor: Shielding Unlearnable Examples Against Data Augmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026). https://doi.org/10.1109/TPAMI.2026.3652456

[10] Hanzhong Guo, Shen Nie, Chao Du, Tianyu Pang, Hao Sun, and Chongxuan Li. 2024. Real-time Identity Defenses against Malicious Personalization of Diffusion Models. arXiv:2412.09844. arXiv:2412.09844 http://arxiv.org/abs/2412.09844v1

[11] Jiaming He, Jiashu Zhang, Guanyu Hou, Shuhan Ye, Hanwei Zhu, Yi Yu, and Xudong Jiang. 2026. ChronoLock: Protecting Videos from Unauthorized Text-to-Video Personalization. arXiv:2606.21146. arXiv:2606.21146 http://arxiv.org/abs/2606.21146v1

[12] Hanxun Huang, Xingjun Ma, Sarah Monazam Erfani, James Bailey, and Yisen Wang. 2022. Unlearnable Examples: Making Personal Data Unexploitable. In *International Conference on Learning Representations (ICLR)*.

[13] Yuxin Huang, Ziming Hong, Mingming Gong, Wanyu Wang, Jing Zhang, and Tongliang Liu. 2026. Delving into the Temporal Challenges of Unified Video Protection Against Image-to-Video and Fine-Tuning-based Customization. arXiv:2607.13336. arXiv:2607.13336 http://arxiv.org/abs/2607.13336v1

[14] Sangwon Jang, June Suk Choi, Jaehyeong Jo, Kimin Lee, and Sung Ju Hwang. 2025. Silent Branding Attack: Trigger-free Data Poisoning Attack on Text-to-Image Diffusion Models. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 8203–8212. https://doi.org/10.1109/CVPR52734.2025.00768

[15] Jaehwan Jeong, Sumin In, Sieun Kim, Hannie Shin, Jongheon Jeong, Sang Ho Yoon, Jaewook Chung, and Sangpil Kim. 2025. FaceShield: Defending Facial Image Against Deepfake Threats. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 10364–10374. https://doi.org/10.1109/ICCV51701.2025.00965

[16] Jun Jia, Hongyi Miao, Yingjie Zhou, Linhan Cao, Yanwei Jiang, Wangqiu Zhou, Dandan Zhu, Hua Yang, Wei Sun, Xiongkuo Min, and Guangtao Zhai. 2025. DLADiff: A Dual-Layer Defense Framework against Fine-Tuning and Zero-Shot Customization of Diffusion Models. arXiv:2511.19910. arXiv:2511.19910 http://arxiv.org/abs/2511.19910v1

[17] Wan Jiang, Yunfeng Diao, He Wang, Jianxin Sun, Meng Wang, and Richang Hong. 2023. Unlearnable Examples Give a False Sense of Security: Piercing through Unexploitable Data with Learnable Examples. In *Proceedings of the 31st ACM International Conference on Multimedia (ACM MM)*. ACM, 8910–8921. https://doi.org/10.1145/3581783.3611833

[18] Jinsu Kim, Yunhun Nam, Minseon Kim, Sangpil Kim, and Jongheon Jeong. 2025. BlurGuard: A Simple Approach for Robustifying Image Protection Against AI-Powered Editing. arXiv:2511.00143. arXiv:2511.00143 http://arxiv.org/abs/2511.00143v1

[19] Nupur Kumari, Bingliang Zhang, Richard Zhang, Eli Shechtman, and Jun-Yan Zhu. 2023. Multi-Concept Customization of Text-to-Image Diffusion. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 1931–1941. https://doi.org/10.1109/CVPR52729.2023.00192

[20] Thanh Van Le, Hao Phung, Thuan Hoang Nguyen, Quan Dao, Ngoc N. Tran, and Anh Tran. 2023. Anti-DreamBooth: Protecting users from personalized text-to-image synthesis. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2116–2127. https://doi.org/10.1109/ICCV51070.2023.00202

[21] Chung Peng Lee, Rachel Hong, Harry H. Jiang, Aster Plotnik, William Agnew, and Jamie Morgenstern. 2026. How Do Data Owners Say No? A Case Study of Data Consent Mechanisms in Web-Scraped Vision-Language AI Training Datasets. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 40. 38808–38816. https://doi.org/10.1609/aaai.v40i45.41225

[22] Jiahao Li, Yiqiang Chen, Yunbing Xing, Yang Gu, and Xiangyuan Lan. 2025. A Survey on Unlearnable Data. arXiv:2503.23536. arXiv:2503.23536 http://arxiv.org/abs/2503.23536v3

[23] Minghui Li, Xianlong Wang, Zhifei Yu, Shengshan Hu, Ziqi Zhou, Longling Zhang, and Leo Yu Zhang. 2025. Detecting and Corrupting Convolution-based Unlearnable Examples. *Proceedings of the AAAI Conference on Artificial Intelligence* 39, 17 (2025), 18403–18411. https://doi.org/10.1609/aaai.v39i17.34025

[24] Zheng Li, Liangbin Xie, Jiantao Zhou, Xintao Wang, Haiwei Wu, and Jinyu Tian. 2025. Anti-Diffusion: Preventing Abuse of Modifications of Diffusion-Based Models. *Proceedings of the AAAI Conference on Artificial Intelligence* 39, 10 (2025), 10582–10590. https://doi.org/10.1609/aaai.v39i10.33149

[25] Yixin Liu, Chenrui Fan, Yutong Dai, Xun Chen, Pan Zhou, and Lichao Sun. 2024. MetaCloak: Preventing Unauthorized Subject-Driven Text-to-Image Diffusion-Based Synthesis via Meta-Learning. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 24219–24228. https://doi.org/10.1109/CVPR52733.2024.02286

[26] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. 2015. Deep Learning Face Attributes in the Wild. In *2015 IEEE International Conference on Computer Vision (ICCV)*. IEEE, 3730–3738. https://doi.org/10.1109/ICCV.2015.425

[27] Hengrui Lou, Zunlei Feng, Jinsong Geng, Erteng Liu, Lechao Cheng, Jie Lei, Jie Song, Mingli Song, and Yijun Bei. 2025. A Large-scale Universal Evaluation Benchmark For Face Forgery Detection. In *Proceedings of the 33rd ACM International Conference on Multimedia*. ACM, 12592–12599. https://doi.org/10.1145/3746027.3758194

[28] Tarik Can Ozden, Ozgur Kara, Oguzhan Akcin, Kerem Zaman, Shashank Srivastava, Sandeep P. Chinchali, and James M. Rehg. 2024. DiffVax: Optimization-Free Image Immunization Against Diffusion-Based Editing. arXiv:2411.17957. arXiv:2411.17957 http://arxiv.org/abs/2411.17957v2

[29] Gan Pei, Jiangning Zhang, Menghan Hu, Zhenyu Zhang, Chengjie Wang, Yunsheng Wu, Guangtao Zhai, Jian Yang, and Dacheng Tao. 2026. Deepfake Generation and Detection: A Benchmark and Survey. *Comput. Surveys* 58, 11 (2026), 1–41. https://doi.org/10.1145/3801962

[30] Sen Peng, Mingyue Wang, Jianfei He, Jijia Yang, and Xiaohua Jia. 2025. CAT: Contrastive Adversarial Training for Evaluating the Robustness of Protective Perturbations in Latent Diffusion Models. arXiv:2502.07225. arXiv:2502.07225 http://arxiv.org/abs/2502.07225v2

[31] Tianrui Qin, Xitong Gao, Juanjuan Zhao, Kejiang Ye, and Cheng-Zhong Xu. 2024. Learning the Unlearnable: Adversarial Augmentations Suppress Unlearnable Example Attacks. In *International Conference on Learning Representations (ICLR)*.

[32] Chunlin Qiu, Ang Li, Tianxiao Huang, Ruilin Gan, Yunjie Ge, Shenyi Zhang, Huayi Duan, Lingchen Zhao, Chao Shen, and Qian Wang. 2026. VOID: Defeating Unauthorized Mimicry in Latent Diffusion Models. arXiv:2606.12263. arXiv:2606.12263 http://arxiv.org/abs/2606.12263v2

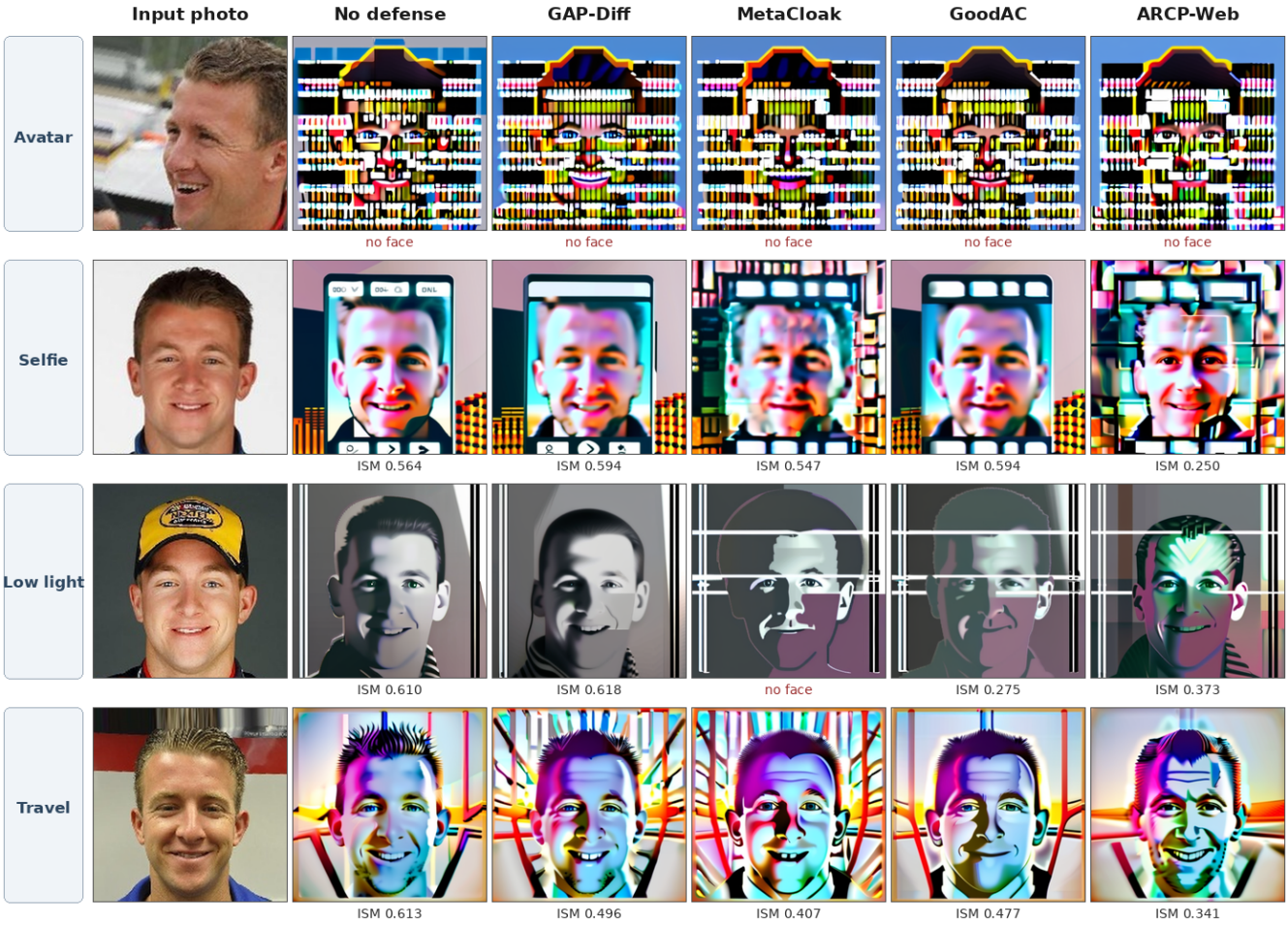



**Figure 5: InstantID outputs after JPEG Q70 (T1). Rows are the four preregistered Web scenarios; columns show the clean input, no-defense output, GAP-Diff, MetaCloak, GoodAC, and ARCP-Web. Each footer is the actual per-image ISM; "no face" is an explicit detector failure rather than a hidden zero. The panel is generated automatically from the same sample ledgers used for Tables 4–5, without cherry-picking identities or seeds.**


[33] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 10674–10685. https://doi.org/10.1109/cvpr52688.2022.01042

[34] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. 2023. DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 22500–22510. https://doi.org/10.1109/CVPR52729.2023.02155

[35] Ali Salar, Qing Liu, Yingli Tian, and Guoying Zhao. 2025. Enhancing Facial Privacy Protection via Weakening Diffusion Purification. arXiv:2503.10350. arXiv:2503.10350 http://arxiv.org/abs/2503.10350v1

[36] Hadi Salman, Alaa Khaddaj, Guillaume Leclerc, Andrew Ilyas, and Aleksander Madry. 2023. Raising the Cost of Malicious AI-Powered Image Editing. In *2023 IEEE Symposium on Security and Privacy (S&P)*.

[37] Pedro Sandoval-Segura, Vasu Singla, Jonas Geiping, Micah Goldblum, Tom Goldstein, and David W. Jacobs. 2022. Autoregressive Perturbations for Data Poisoning. In *Advances in Neural Information Processing Systems (NeurIPS)*. Neural Information Processing Systems Foundation, 27374–27386. https://doi.org/10.52202/068431-1985

[38] Shawn Shan, Jenna Cryan, Emily Wenger, Haitao Zheng, Rana Hanocka, and Ben Y. Zhao. 2023. Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models. In *32nd USENIX Security Symposium (USENIX Security 23)*.

[39] Shawn Shan, Emily Wenger, Jiayun Zhang, Huiying Li, Haitao Zheng, and Ben Y. Zhao. 2020. Fawkes: Protecting Privacy against Unauthorized Deep Learning Models. USENIX Security Symposium 2020. arXiv:2002.08327 http://arxiv.org/abs/2002.08327v2

[40] Jee-Woo Shin, Chayapol Kamyod, and Chung-Pyo Hong. 2026. Latent Diffusion Models: A Survey on Foundations, Variants, and Web-scale Deployments. *Journal of Web Engineering* (2026). https://doi.org/10.13052/jwe1540-9589.2546

[41] Yiren Song, Pei Yang, Hai Ci, and Mike Zheng Shou. 2025. IDProtector: An Adversarial Noise Encoder to Protect Against ID-Preserving Image Generation. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 3019–3028. https://doi.org/10.1109/CVPR52734.2025.00287

[42] Matteo Trippodo, Federico Becattini, and Lorenzo Seidenari. 2025. Immunizing Images from Text to Image Editing via Adversarial Cross-Attention. arXiv:2509.10359. arXiv:2509.10359 http://arxiv.org/abs/2509.10359v1

[43] Derui Wang, Minhui Xue, Bo Li, Seyit Camtepe, and Liming Zhu. 2025. Provably Unlearnable Data Examples. In *Proceedings of the 2025 Network and Distributed System Security Symposium (NDSS)*. Internet Society. https://doi.org/10.14722/ndss.2025.240886

[44] Guanyu Wang, Kailong Wang, Yihao Huang, Mingyi Zhou, Geguang Pu, and Li Li. 2026. Privacy Protection Against Personalized Text-to-Image Synthesis via Cross-image Consistency Constraints. arXiv:2504.12747. In *Proceedings of the 2026 International Conference on Multimedia Retrieval (ICMR)*. ACM, 2003–2011. https://doi.org/10.1145/3805622.3810856 arXiv:2504.12747

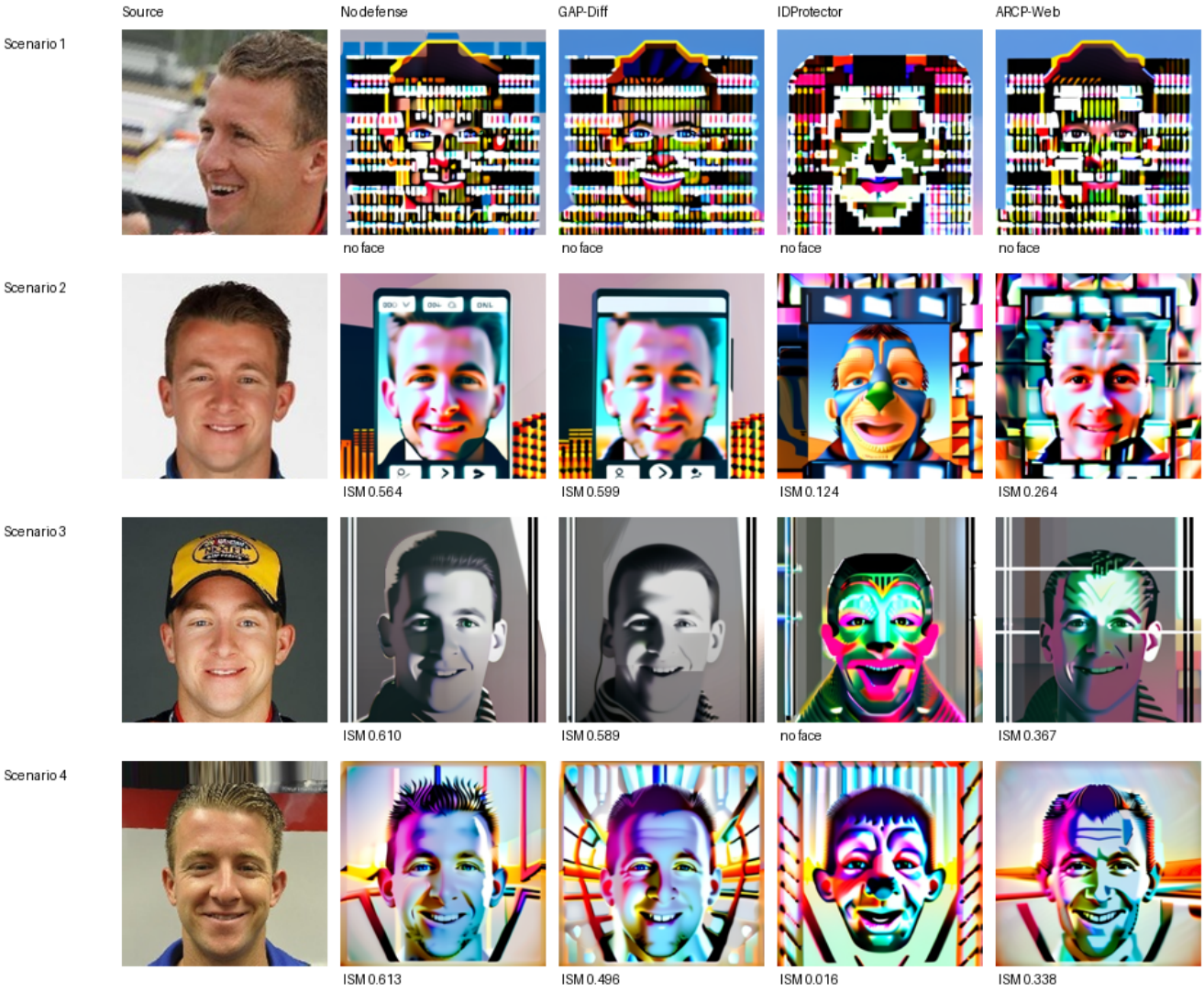



**Figure 6: Preregistered qualitative panel for the first manifest identity at T1 (JPEG Q70) under InstantID. Rows are the four fixed Web scenarios. Images and footer ISM values are read directly from the per-sample ledger; no identity, seed, or output is selected by outcome.**


[45] L.F.S. Wang, Qianyue Hu, Wei Lu, and Xiangyang Luo. 2026. Safeguarding Facial Identity against Diffusion-based Face Swapping via Cascading Pathway Disruption. arXiv:2601.14738. arXiv:2601.14738 http://arxiv.org/abs/2601.14738v1

[46] Qixun Wang, Xu Bai, Haofan Wang, Zekui Qin, Anthony Chen, Huaxia Li, Xu Tang, and Yao Hu. 2024. InstantID: Zero-shot Identity-Preserving Generation in Seconds. arXiv:2401.07519. arXiv:2401.07519 http://arxiv.org/abs/2401.07519v2

[47] Songping Wang, Yueming Lyu, Shiqi Liu, Ning Li, Tong Tong, Hao Sun, and Caifeng Shan. 2025. Anti-Aesthetics: Protecting Facial Privacy against Customized Text-to-Image Synthesis. arXiv:2504.12129. arXiv:2504.12129 http://arxiv.org/abs/2504.12129v2

[48] Songping Wang, Yueming Lyu, Shiqi Liu, Chen Zhao, Ziyuan Chen, Ning Li, Jing Dong, and Caifeng Shan. 2026. Hierarchical Anti-Aesthetics: Protecting Facial Privacy against Customized Diffusion Models. arXiv:2607.02038. arXiv:2607.02038 http://arxiv.org/abs/2607.02038v2

[49] Tao Wang, Yushu Zhang, Xiangli Xiao, Kun Xu, Lin Yuan, Wenying Wen, and Yuming Fang. 2025. Make Identity Unextractable yet Perceptible: Synthesis-Based Privacy Protection for Subject Faces in Photos. arXiv:2509.11249. arXiv:2509.11249 http://arxiv.org/abs/2509.11249v1

[50] Yuxiang Wei, Yiheng Zheng, Yabo Zhang, Ming Liu, Zhilong Ji, Lei Zhang, and Wangmeng Zuo. 2025. Personalized Image Generation with Deep Generative Models: A Decade Survey. arXiv:2502.13081. arXiv:2502.13081 http://arxiv.org/abs/2502.13081v1

[51] Qiangqiang Wu, Yi Yu, Chenqi Kong, Ziquan Liu, Jia Wan, Haoliang Li, Alex C. Kot, and Antoni B. Chan. 2025. Temporal Unlearnable Examples: Preventing Personal Video Data from Unauthorized Exploitation by Object Tracking. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 11110–11121. https://doi.org/10.1109/ICCV51701.2025.01034

[52] Stanley Wu, Ronik Bhaskar, Anna Yoo Jeong Ha, Shawn Shan, Haitao Zheng, and Ben Y. Zhao. 2025. On the Feasibility of Poisoning Text-to-Image AI Models via Adversarial Mislabeling. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2848–2862. https://doi.org/10.1145/3719027.3744845

[53] Zhihao Wu, Yushi Cheng, Tianyang Sun, Xiaoyu Ji, and Wenyuan Xu. 2025. MYOPIA: Protecting Face Privacy from Malicious Personalized Text-to-Image Synthesis via Unlearnable Examples. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 39. 905–913. https://doi.org/10.1609/aaai.v39i1.

32075
[54] Long Ya Xu, Jiakai Wang, Haojie Hao, Haotong Qin, Jiejie Zhao, and Xianglong Liu. 2025. Harnessing Global-Local Collaborative Adversarial Perturbation for Anti-Customization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 13414–13423. https://doi.org/10.1109/CVPR52734.2025.01252
[55] Zachary Yahn, Fatih Ilhan, Tiansheng Huang, Selim Tekin, Sihao Hu, Yichang Xu, Margaret Loper, and Ling Liu. 2026. Personalized Face Privacy Protection From a Single Image. arXiv:2605.19032. arXiv:2605.19032 http://arxiv.org/abs/2605.19032v1
[56] Jing Yang, Runping Xi, Yingxin Lai, Xun Lin, and Zitong Yu. 2024. DDAP: Dual-Domain Anti-Personalization against Text-to-Image Diffusion Models. *IEEE International Joint Conference on Biometrics (IJCB)* (2024), 1–10. https://doi.org/10.1109/IJCB62174.2024.10744490
[57] Wenkui Yang, Jie Cao, Junxian Duan, and Ran He. 2025. Towards Robust Defense against Customization via Protective Perturbation Resistant to Diffusion-based Purification. arXiv:2509.13922. arXiv:2509.13922 http://arxiv.org/abs/2509.13922v2
[58] Hu Ye, Jun Zhang, Sibo Liu, Xiao Han, and Wei Yang. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. arXiv:2308.06721. arXiv:2308.06721 http://arxiv.org/abs/2308.06721v1
[59] Mengying Zhang, Derui Wang, Ruoxi Sun, Xiaoyu Xia, Shuang Hao, and Minhui Xue. 2026. SoK: Unlearnability and Unlearning for Model Dememorization. arXiv:2605.11592. arXiv:2605.11592 http://arxiv.org/abs/2605.11592v1
[60] Haotian Zhu, Shuchao Pang, Zhigang Lu, Yongbin Zhou, and Minhui Xue. 2025. GAP-Diff: Protecting JPEG-Compressed Images from Diffusion-based Facial Customization. In *Network and Distributed System Security Symposium (NDSS)*. https://doi.org/10.14722/ndss.2025.242088

# A REPRODUCIBILITY AND ETHICS

## A.1 Artifact map

The cohort manifest and audit are stored with the released `data/web40` artifact. The frozen experiment configuration is `configs/web40_single_intervention.yaml`. The four-identity entry points are the three slurm launchers `slurm/web4_eval.sbatch`, `slurm/web4_metacloak_eval.sbatch`, and `slurm/web4_summary.sbatch`. Protection is generated by `scripts/run_official_web40_protector.py` and the ARCP-v4 scripts. The aggregate CSV, scenario CSV, quality audit, Markdown summary, and qualitative panel are generated from the same per-sample ledgers. Each runner can be resumed without overwriting completed cells.

## A.2 Integrity rules

Legacy 37-identity results, failed ingestion attempts, and results from the old cumulative funnel are excluded from the new tables. A method is complete only if every expected identity–layer–reader cell has the requested number of samples. NaNs, missing faces, and failed jobs remain explicit. Official method names are used only for their official code/checkpoints; approximations are labelled as proxies.

## A.3 Ethics

The study evaluates defenses using public research datasets and released attack pipelines. Held-out references are used only for aggregate scoring. Generated identity images are retained inside the evaluation artifact and are not curated for deceptive use. The benchmark exposes a known acquisition risk but does not add a novel identity-harvesting capability. Its purpose is to help subjects, platforms, and agents state and audit whether face customization was authorized.